Experiment 11

Date of performance: _____

Date of submission: _____

Title : To Create an Automated Customer Support Triage System Using n8n

Aim: To design and develop an automated Customer Support Triage System using n8n and an HTML frontend that accepts customer support requests, analyzes the issue and message details, determines the priority and category, assigns the appropriate support action, and displays the triage result.

Software Requirement

- Node.js
- n8n
- Google Chrome or any web browser
- HTML
- CSS
- JavaScript

Theory

n8n is a workflow automation platform that allows users to connect different applications, APIs, and services using visual workflow nodes.

Customer support triage is a technique used to determine how a customer request should be handled based on its content, urgency, and type. Instead of manually checking every support request, the system analyzes the request and generates a support recommendation.

In this experiment, the HTML frontend is used to enter customer support information such as:

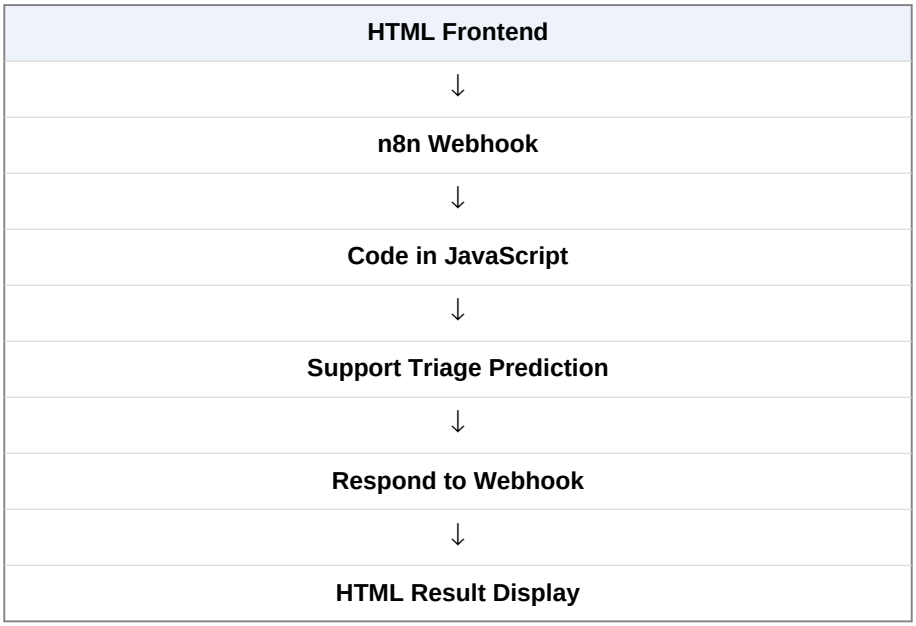
- Customer ID
- Issue Category
- Customer Message
- Order / Ticket ID

The entered data is sent to the n8n Webhook using a POST request. The Code in JavaScript node processes the received support data. It checks the message and category values and determines the priority level of the request. The system uses the following conditions:

- High Priority: Message contains urgent, payment, fraud, or account access issues
- Medium Priority: Product, order, delivery, or service issues requiring support
- Low Priority: General questions or informational requests

Based on the priority level, the system generates a support action and assigns a suitable support team. The Respond to Webhook node sends the processed result back to the HTML frontend, where the triage result is displayed.

Workflow Diagram



Algorithm

1. Start the HTML frontend.
2. Enter the Customer ID.
3. Enter the issue category.
4. Enter the customer support message.
5. Enter the Order / Ticket ID.
6. Send the support data to the n8n Webhook using a POST request.
7. Receive the support data in the Webhook node.
8. Pass the received data to the JavaScript Code node.
9. Check the customer message and issue category.
10. Determine the request priority as High, Medium, or Low.
11. Generate the appropriate support action.
12. Assign the request to the suitable support team.
13. Send the generated result through the Respond to Webhook node.
14. Display the customer support triage result on the HTML webpage.
15. Stop.

Implementation

1. HTML Frontend

The HTML webpage contains input fields for:

Customer ID, Issue Category, Customer Message, Order / Ticket ID

The user enters the customer support information and clicks the **Analyze Request** button. The data is converted into JSON format and sent to the n8n Webhook.

Example Input

Customer ID: C001 , Issue Category: Payment , Customer Message: My payment was charged twice and I need urgent help , Order / Ticket ID: T1001

2. n8n Webhook

The Webhook node receives the customer support information sent by the HTML frontend. The request is sent using the POST method.

Example data:

```
{
    "customer_id": "C001",
    "issue_category": "Payment",
    "message": "My payment was charged twice and I need urgent help",
    "ticket_id": "T1001"
}
```

3. Respond to Webhook

The Respond to Webhook node sends the processed output from the Code node back to the HTML frontend. The HTML webpage then displays the received customer support triage result.

Output:

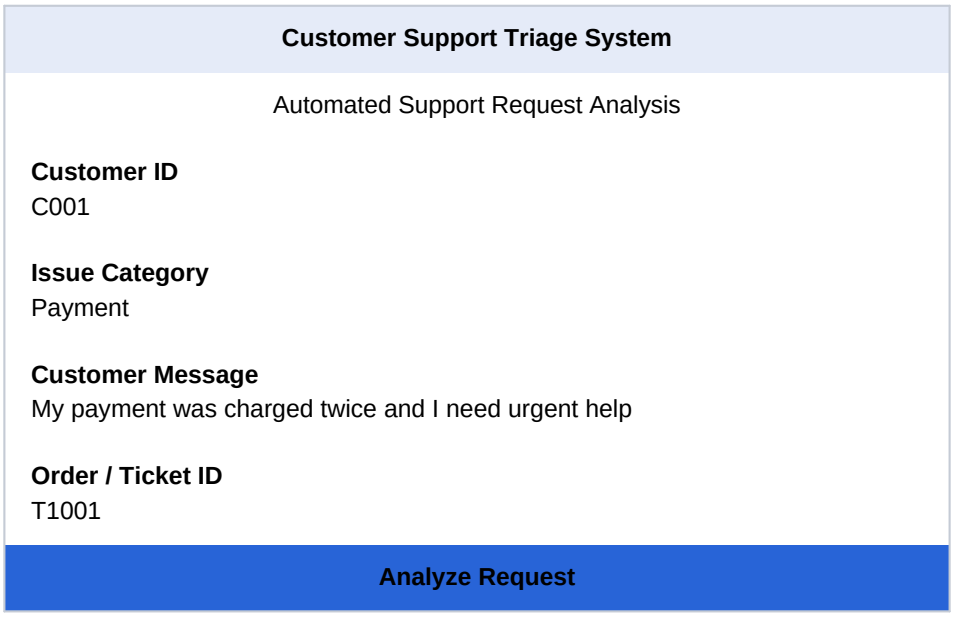


Figure 1: Customer Support Triage System HTML interface

Figure 2: n8n Customer Support Triage workflow

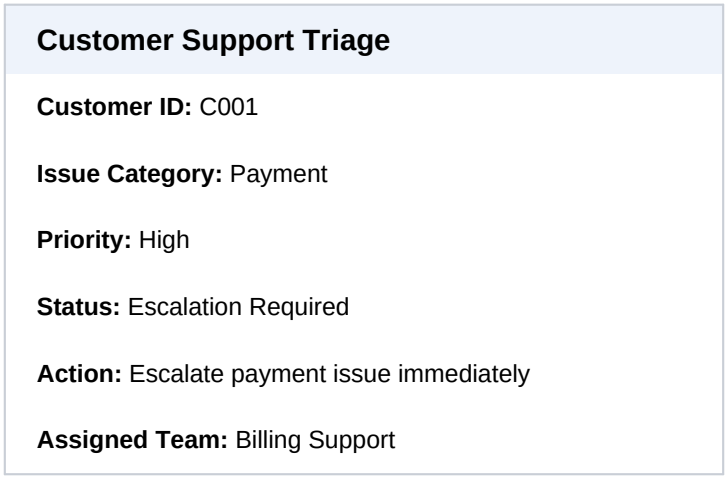
The workflow receives the support request through the Webhook, processes the customer message and issue category using the JavaScript Code node, and returns the generated triage result through the Respond to Webhook node.

Figure 3: Customer data processed by Code node

customer_id	issue_category	priority	status	team	action
C001	Payment	High	Escalation Required	Billing Support	Escalate payment issue immediately

The Code node classifies the request as **High** priority because the example message contains an urgent payment issue.

Figure 4: Customer support triage result displayed on HTML webpage



Result

The **Automated Customer Support Triage System** was successfully developed using n8n, JavaScript, and HTML. The workflow successfully accepted customer support data through the Webhook, processed the issue category and customer message, determined the request priority, generated a support recommendation, assigned the appropriate support team, and displayed the final result on the HTML webpage.

Conclusion: The experiment demonstrated how n8n can be used to create an automated customer support triage workflow. The HTML frontend accepted customer support information, the Webhook received the data, the JavaScript Code node analyzed the support request, and the Respond to Webhook node returned the triage result to the webpage. The workflow can be reused with different customer requests by changing the input data and executing the analysis again.

Marks & Signature

R1 (3)	R2 (5)	R3 (5)	R4 (2)	Total Marks (15)	Signature